

From topological principle to a measurable cell-state statistic

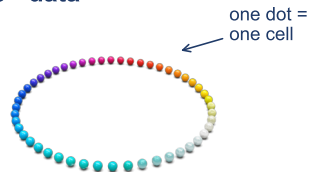
(a) topology counts holes $\rightarrow$ (b) scRNA-seq cells form one such loop $\rightarrow$ (c) persistent homology measures its lifespan

a · principle



Same hole, different geometry

b · data



Cells trace a state-space loop



measure with persistent homology

c · measurement

How long the loop survives
= this paper's $\max H_1$ persistence

